Control Reinforcement Learning

An Exploration and Evaluation on LegalBench

Garrett McNeill, Kyle Spurlock, Nicholas Johnson, Olfa Nasraoui

UNIVERSITY OF LOUISVILLE®

Introduction

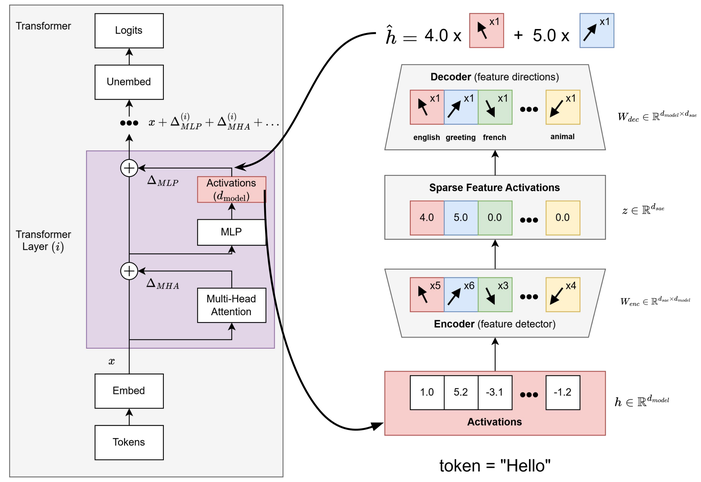
- Neurons in large language models are often **polysemantic**, encoding **multiple** unrelated concepts and making model behavior **difficult to interpret**.
- Bricken et al. [1] used sparse autoencoders (SAEs) to decompose model activations into **sparse**, more **monosemantic features** using dictionary learning.
- The resulting **interpretable features** enable better understanding of model internals and support **mechanistic interpretability**, circuit analysis, and feature-level model **control**.
- **However**, SAEs reveal **which** features activate for particular inputs, but do **not** reliably indicate which features **causally affect** the model's **output**.
- **Control Reinforcement Learning (CRL)** [2] trains a reinforcement learning agent to **select** features that **change** the model's output.
  - Agent **reward**: based on **evaluation** of the modified output during training **against a benchmark**.
  - CRL's feature-selection behavior and task performance can vary based on implementation and training choices, including hyperparameters and reward functions.
  - CRL can identify **subsets of features** that are **semantically meaningful to the task** on which it was trained.

Research Goals

- This REU conducted an exploratory research to understand and replicate on a yet untested task (LegalBench) a recently proposed state of the art methodology, called Control Reinforcement Learning (CRL).

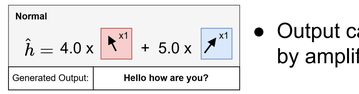
- The experiments attempted to answer the following questions:
  - Can CRL be replicated on a task outside of the task set used in the original paper?
  - Will CRL chose semantically meaningful features related to the tasks it gets trained on?
  - How robust is the CRL method, and do changes in the reward function and hyperparameters effect feature selection and model performance?
  - Does CRL affect overall LLM performance?

Methodology and Evaluation



- Each sublayer in a Transformer produces a residual **Δ**.
- Gradually transform a token representation ***x*** into a state relevant for predicting next possible tokens.

- An SAE hooks into the activations of the MLP in a Transformer layer.
- The objective is to represent residual stream activations as a sparse linear combination of learned features at the hooked layer.



- Output can be **steered** by amplifying features.

- CRL learns an RL agent that **selects** which features to amplify to allow for **interpretable fine-tuning** on a task.
- Uses Advantage Actor-Critic (A2C) trained with Proximal Policy Optimization (PPO).
  - **State** $\mathscr{S}$: $\mathbf{s} = \mathbf{x} \in \mathbb{R}^d$, residual stream activation at layer ℓ.
  - **Action** $\mathscr{A}$: $\mathbf{a} \in \{0,1\}^{\text{dsae}}$, binary feature selection vector.
  - **Transition** $\mathscr{P}$: Transformer forward pass with feature steering.
  - **Reward** $\mathscr{R}$: Task-specific reward from output quality.

$$\boldsymbol{\mu} = \pi_\theta(\mathbf{s})$$
$$\mathbf{a} = \mathrm{TopK}(\boldsymbol{\mu}, k)$$
$$\tilde{\mathbf{x}}_t = \mathbf{x}_t + c \cdot \mathbf{a}_t \mathbf{W}_{dec}$$
$$V_\phi(\mathbf{s}) = \mathbb{E}_{\pi_\theta}[r \mid \mathbf{s}]$$

- Policy $\pi_\theta : \mathbb{R}^d \to \mathbb{R}^{d_{dict}}$. Logits are sampled and the Top-1 is selected.
- $\tilde{\mathbf{x}}_t$ is the steered activation with ***c*** steering coeff.
- Critic $V_\phi : \mathbb{R}^d \to \mathbb{R}$ estimates reward (*success or not*) from residual.

Model / Benchmark Details

- Used IssueSpotting task from LegalBench, specifically only learned hands tasks.
- Randomly shuffled learned hands tasks test set into a train, validate, test set for our agent.
- Binary rewards, standard was +1 for a correct answer and 0 for a incorrect answer.
  - Some test runs changed 0 to -10 for incorrect answer.
- Used Gemma 2-2B (not instruction tuned).
- Applied GemmaScope 2 SAE at layer 20, with width of 16k.
- Actor and Critic both 2 layer MLP with TanH activation.
- Trained for 4000 steps.
- Evaluation batch size of 64.
- Clip epsilon set to 0.2 for PPO.
- Base learning rate set to 5e-5.
- Tested different learning rates, reward functions, and entropy added to actor loss.

Comparison of CRL Implementation Variants

Test Run	Mean Score (%)	Feature Diversity
Baseline	48.33 ± 1.77	–
Initial	51.67 ± 0.72	3.66 ± 0.94
$r = -10$	51.45 ± 1.49	7.00 ± 0.82
$lr =$ 5e-6	51.78 ± 0.63	16.33 ± 1.70
$lr =$ 5e-6 and $r = -10$	50.33 ± 1.51	26.67 ± 0.94
$lr =$ 5e-6 and entropy for actor loss	50.89 ± 1.64	12.66 ± 0.94

Feature Number	Semantic Meaning
1891	occurrences of the word "first" and its related phrases signifying **initial** achievements or events
6627	prominent **names** and **references** related to social issues or health crises
8170	terms related to **security** and **privacy** in data handling
14774	**technical** terms and structured **data formats**

Conclusions and Future Work

- CRL was replicable on tasks that were not included in the initial papers test set, even with slightly different methodology,
  - This shows the method's robustness.
  - Replicability can be verified by examining the model's feature selection during test phases
- Different learning rates, the addition of entropy, and different reward functions all affect the agents feature selection diversity
  - sometimes changing the learned policy and feature selection itself
- Different methods of CRL had slight effect on the model's performance on issue spotting tasks from LegalBench.
  - Affecting the mean scores and standard deviation of performance
- Future work includes trying different SAE layers, different hyperparameters, expanding test sets, different tasks, different LLMs, evaluating cross task performance, multi agent RL on multiple layers, and multi objective RL on multiple tasks

Acknowledgement

This research was supported by the U.S. National Science Foundation (NSF) under grant CNS-2349076.

References

[1] Bricken, T., Templeton, A., Batson, J., Chen, B., Jermyn, A., Conerly, T., Turner, N., Anil, C., Denison, C., Askell, A., Lasenby, R., Wu, Y., Kravec, S., Schiefer, N., Maxwell, T., Joseph, N., Hatfield-Dodds, Z., Tamkin, A., Nguyen, K., McLean, B., Burke, J. E., Hume, T., Carter, S., Henighan, T., & Olah, C. (2023). Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread. https://transformer-circuits.pub/2023/monosemantic-features/index.html

[2] S. Cho, Z. Wu, and A. Koshiyama, "Control Reinforcement Learning: Interpretable Token-Level Steering of LLMs via Sparse Autoencoder Features," arXiv preprint arXiv:2602.10437, 2026.

[3] R. Huben, H. Cunningham, L. R. Smith, A. Ewart, and L. Sharkey, "Sparse Autoencoders Find Highly Interpretable Features in Language Models," in International Conference on Learning Representations (ICLR), 2024